PHYSICS 101

INTRODUCTION TO PHYSICS

- ❑ **CENTER OF MASS**
- ❑ **ROTATIONAL MOTION**
- ❑ **TORGUE**
- ❑ **VECTOR PRODUCT**
- ❑ **TORGUE**
- ❑ **MOMENT OF INERTIAL**
- ❑ **ANGULAR MOMENTUM**

Center of Mass Formula

What is the Center of Mass?

The center of mass of a body or system of a particle is defined as a point where the whole of the mass of the body or all the masses of a set of particles appeared to be concentrated.

When we are studying the dynamics of the motion of the system of a particle as a whole, then we need not worry about the dynamics of individual particles of the system. Rather than we have to focus on the dynamic of a unique point corresponding to that system.

The motion of this unique point is similar to the motion of a single particle whose mass is equal to the sum of all individual particles of the system. Also the resultant of all the forces exerted on all the particles of the system by surrounding bodies is exerted directly to that particle. This point is known as the center of mass of the system of particles.

Centre of Mass Formula

We can extend the formula for the center of mass to a system of particles. We can apply the equation individually to each axis also.

Although the center of mass and the center of gravity often coincide, these are all different concepts. Meanwhile, the center of gravity and the center of mass are only equal when the entire system is subject to a uniform gravitational field.

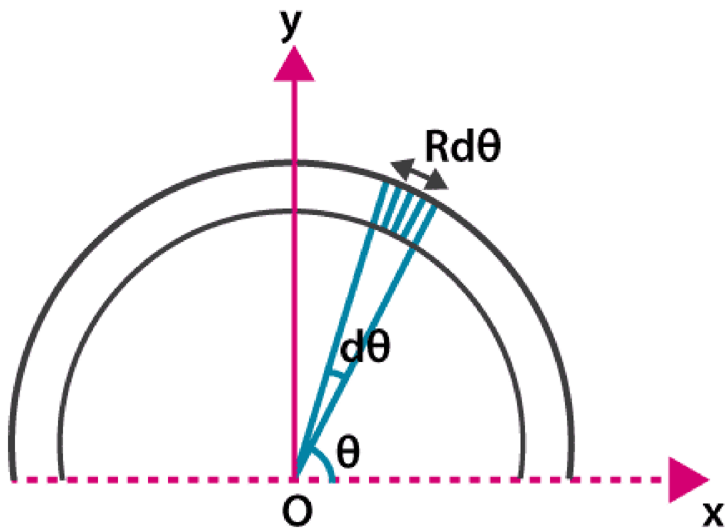
$$x_{cm} = \frac{\sum x_i m_i}{\sum m_i}, \quad y_{cm} = \frac{\sum x_i m_i}{\sum m_i}, \quad z_{cm} \frac{\sum x_i m_i}{\sum m_i}$$

We can use the above formula if we have pointed objects. On the other hand, we have to take a different approach if we have to find the center of mass of an extended object like a rod. Then we will consider a differential mass and its position and then integrate it over the entire length.

- $X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{60 \times 0.0 + 15 \times 0.10}{60 + 15} = \frac{1.50}{75} - 0.02\text{m}$

The center of mass will be at 0.020 m from the circle.

Centre of mass for a semi-circular ring of radius (R) and mass (M)



Consider a differential element of length (dl) of the ring whose radius vector makes an angle θ with the x-axis. If the angle subtended by the length (dl) is dθ at the centre, then

$dl = Rd\theta$ 1

Then the mass of the element is dm,

$dm = \lambda Rd\theta$ 2

since

$x_{cm} = \frac{1}{M}\int xdm = \frac{1}{M}\int_0^\pi R\cos\ \theta(\lambda Rd\theta) = 0$ 3

$y_{cm} = \frac{1}{M}\int_0^\pi(R\sin\ \theta)(\lambda Rd\theta)$ 4

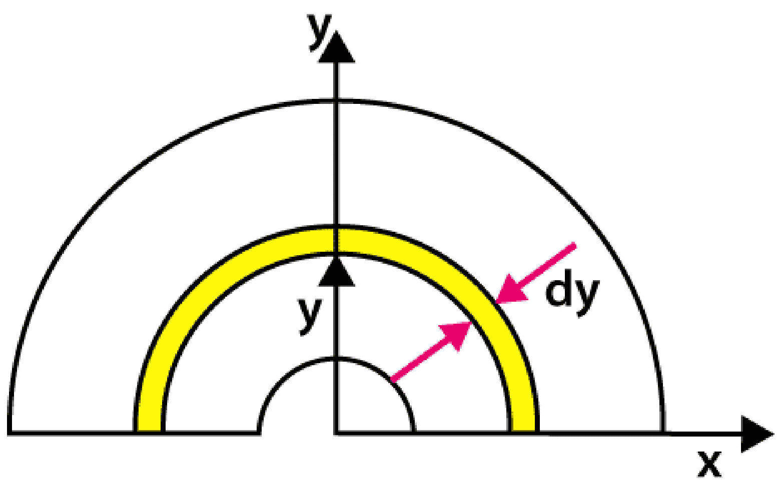
Then equation become

$= \frac{\lambda R^2}{M}\int_0^\pi \sin\ \theta d\theta = \frac{\lambda R^2}{\lambda\pi R}(-\cos\ \theta)_0^\pi$ 5

$= \frac{2R}{\pi}$ 6

Is the final equation to be use for semi circule

Centre of mass of a thin uniform disc of radius (R)



Solution:

The disc is an analogy of a ring. We are considering an elemental ring at y distance from origin and thickness dy.

As we know, $x_{cm} = 0$ (for a ring)

$y_{cm} = \frac{2R}{\pi}$ (for a ring)

Now for disc

The disc is a combination of N number of elemental rings.

y_{cm} for a disc is

$$y_{cm} = \frac{1}{M}\int_0^R dmy$$

$$dm = \frac{M}{A}da = \frac{M}{\frac{\pi R^2}{2}}\pi r dr$$

$$y_{cm} = \frac{1}{M}\int_0^R \left(\frac{M}{\frac{\pi R^2}{2}}(\pi r dr)\right)\frac{2r}{\pi}$$

$$y_{cm} = \frac{4}{\pi R^2}\int_0^R r^2 dr$$

$$y_{cm} = \frac{4}{\pi R^2}\left[\frac{R^3}{3}\right]$$

$$y_{cm} = \frac{4R}{3\pi}.$$

System of Particles and Centre of Mass

Till now, we have dealt with the translational motion of rigid bodies where a rigid body is also treated as a particle. But, when a rigid body undergoes rotation, all of its constituent particles do not move in an identical fashion. Still, we must treat it as a system of particles in which all the particles are rigidly connected to each other.

On the contrary, we may have particles or bodies not connected rigidly to each other but maybe interact with each other through internal forces. Despite the complex motion of which a system of the particle is capable, there is a single point known as the centre of mass or mass centre whose translational motion is characteristic of the system.

Applications

The centre of mass of an aircraft is a critical position that considerably impacts the aircraft's equilibrium. The centre of mass must lie within certain limits in order for the aircraft to be secure enough to fly safely. Determining the centre of mass's position is significant since it enables you to assess dynamical problems based on the centre of mass's motion. When a hammer is thrown into the air, its centre of mass will travel along a curved path. It's as if a particle had been flung into the air.

Rotational motion

Rotational motion can be defined as the motion of an object around a circular path, in a fixed orbit." The dynamics for rotational motion are completely analogous to linear or translational dynamics. Many of the equations for mechanics of rotating objects are similar to the motion equations for linear motion.

Examples of Rotation about a fixed point

Ceiling fan rotation, rotation of the minute hand and the hour hand in the clock, and the opening and closing of the door are some of the examples of rotation about a fixed point.

Rotational Motion & Work-Energy Theorem

The work-energy theorem for rotation predicts the relationship between the work done by the torque of the force and the change in its kinetic energy when a rigid body rotates about a fixed axis. The amount of work done by all the torques operating on a rigid body under a fixed axis rotation (pure rotation) equals the change in its rotational kinetic energy, according to the work-energy theorem for rotation. The mathematical formula for the work-energy theorem in rotational motion is given below.

$W_{torque} = \Delta KE_{rotational}$

The work done by a torque can be computed using an analogy with work done by force. The dot product of force and displacement of the point of force applied is used to calculate work done by force. In the case of angular motion, torque replaces force and linear displacement is replaced by angular displacement. Then the equation for work becomes,

$W = \int \tau \cdot d\theta$

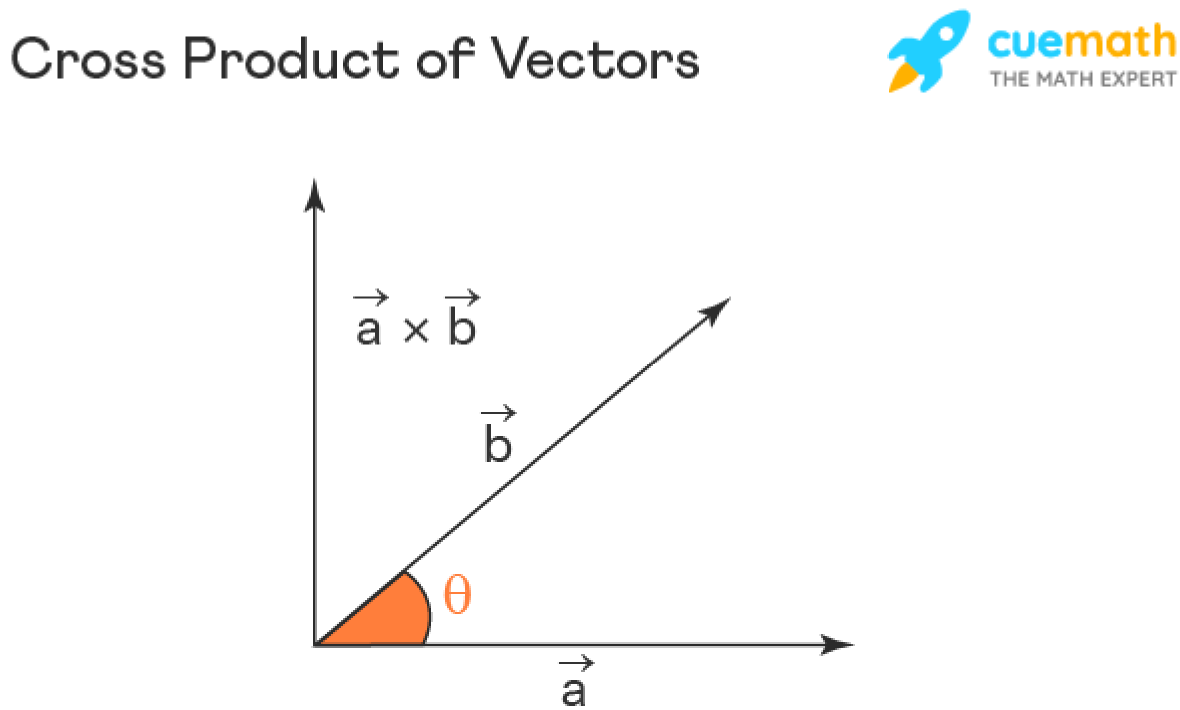
Examples of Rotation about an axis of rotation

Rotation about an axis of rotation includes translational as well as rotational motion. The best example of rotation about an axis of rotation is pushing a ball from an inclined plane. The ball reaches the bottom of the inclined plane through translational motion while the motion of the ball is happening as it is rotating about its axis, which is rotational motion.

Another example of rotation about an axis of rotation is the earth's motion. The earth rotates about its axis every day, and it also rotates around the sun once every year. This is a classic example of translational motion as well as rotational motion.

Torque is the force required to rotate an object around an axis. Just as force causes an object to accelerate in linear **kinematics**, the torque also causes an object to acquire angular acceleration. It is also referred to as the moment, moment of force, rotational force, or turning effect, depending on the field of study. Torque is a vector quantity.

VECTOR PRODUCT



We can understand this with an example that if we have two vectors lying in the X-Y plane, then their cross product will give a resultant vector in the direction of the Z-axis, which is perpendicular to the XY plane. The × symbol is used between the original vectors. The vector product or the cross product of two vectors is shown as:

$\vec{a}\times\vec{b}=\vec{c}$

Here $\vec{a}$ and $\vec{b}$ are two vectors, and $\vec{c}$ is the resultant vector. Let θ be the angle formed between $\vec{a}$ and $\vec{b}$ and $\hat{n}$ is the unit vector perpendicular to the plane containing both $\vec{a}$ and $\vec{b}$

. The cross product of the two vectors is given by the formula:

$\vec{a}\times\vec{b}=|a||b|\sin(\theta)\hat{\ }$

Cross Product

Let us assume that →a and →b are two vectors, such that →a= a1^i+b1^j+c1^k and →b = a2^i+b2^j+c2^k then by using determinants, we could find the cross product and write the result as the cross product formula using the following matrix notation.

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

The cross product of two vectors is also represented using the cross product formula as:

$\vec{a}\times\vec{b}=\hat{i}(b_1c_2-b_2c_1)-\hat{j}(a_1c_2-a_2c_1)+\hat{k}(a_1b_2-a_2b_1)$

Note: $\hat{i},\hat{j}$, and $\hat{k}$

are the unit vectors in the direction of x axis, y-axis, and z -axis respectively.

Properties of Product Of Vectors

The dot product of the unit vector is studied by taking the unit vectors $\hat{i}$ along the x-axis, $\hat{j}$ along the y-axis, and $\hat{k}$ along the z-axis respectively. The dot product of unit vectors $\hat{i}$, $\hat{j}$, $\hat{k}$

follows similar rules as the dot product of vectors. The angle between the same vectors is equal to 0º, and hence their dot product is equal to 1. And the angle between two perpendicular vectors is 90º, and their dot product is equal to 0.

- $\hat{i}.\hat{i}$

$= \hat{j}.\hat{j} = \hat{k}.\hat{k}$

- $= 1$
- $\hat{i}.\hat{j} = \hat{j}.\hat{k} = \hat{k}.\hat{i}$

- $= 0$

The cross product of unit vectors $\hat{i}$

, ^j, ^k

follows similar rules as the cross product of vectors. The angle between the same vectors is equal to 0º, and hence their cross product is equal to 0. And the angle between two perpendicular vectors is 90º, and their cross product gives a vector, which is perpendicular to the two given vectors.

- $\vec{i} \times \vec{i} = \vec{j} \times \vec{j} = \vec{k} \times \vec{k} = 0$

ANGULAR MOMENTUM

Momentum is the product of mass and the velocity of the object. Any object moving with mass possesses momentum. The only difference in angular momentum is that it deals with rotating or spinning objects. So is it the rotational equivalent of linear momentum?

What is Angular Momentum?

If you try to get on a bicycle and balance without a kickstand, you will probably fall off. But once you start pedaling, these wheels pick up angular momentum. They are going to resist change, thereby making balancing gets easier.

Angular momentum is defined as:

The property of any rotating object given by moment of inertia times angular velocity.

It is the property of a rotating body given by the product of the moment of inertia and the angular velocity of the rotating object. It is a **vector quantity**, which implies that the direction is also considered here along with magnitude.

Angular Momentum Formula

Angular momentum can be experienced by an object in two situations. They are:

Point object: The object accelerating around a fixed point. For example, Earth revolving around the sun. Here the angular momentum is given by:

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- **Point object:** The object accelerating around a fixed point. For example, Earth revolving around the sun. Here the angular momentum is given by:

$$\vec{L} = \vec{r} \times \vec{p}$$

Where: $\vec{L}$ is the angular velocity

$\vec{p}$ is the angular momentum

r is the radius (distance between the object and the fixed point about which it revolves)

Extended object: The object, which is rotating about a fixed point. For example, Earth rotates about its axis. Here the angular momentum is given by:

$$\underset{L}{\rightarrow} = \underset{p}{\rightarrow} \times \underset{w}{\rightarrow}$$

Where: $\underset{L}{\rightarrow}$ is the angular momentum

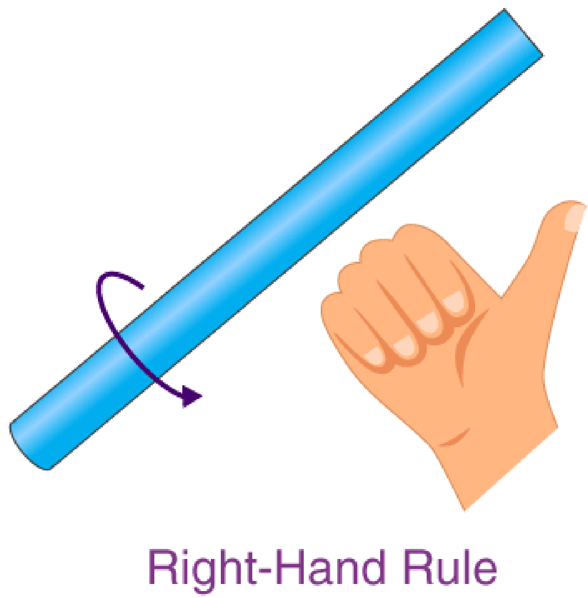
$\underset{w}{\rightarrow}$ is the angular velocity

$\underset{I}{\rightarrow}$ is the rotational inertial

Angular Momentum Quantum Number

Angular momentum quantum number is synonymous with Azimuthal quantum number or secondary quantum number. It is a quantum number of an atomic orbital that decides the angular momentum and describes the size and shape of the orbital. The typical value ranges from **0 to 1.**

Right-Hand Rule



The direction of angular momentum is given by the right-hand rule, which states that:

- If you position your right hand such that the fingers are in the direction of r.
- Then curl them around your palm such that they point towards the direction of Linear momentum(p).
- The outstretched thumb gives the direction of angular momentum(L).

Examples of Angular Momentum

We knowingly or unknowingly come across this property in many instances. Some examples are explained below.

Ice-skater

When an ice-skater goes for a spin she starts off with her hands and legs far apart from the centre of her body. But when she needs more angular velocity to spin, she gets her hands and leg closer to her body. Hence, her angular momentum is conserved, and she spins faster.

Ice-skater

Gyroscope

A gyroscope uses the principle of angular momentum to maintain its orientation. It utilises a spinning wheel that has 3 degrees of freedom.

When it is rotated at high speed it locks on to the orientation, and it won't deviate from its orientation. This is useful in space applications where the attitude of a spacecraft is a really important factor to be controlled.

What is a coordinate system?

A **coordinate system** is a method for identifying the location of a point on the earth. Most coordinate systems use two numbers, a **coordinate**, to identify the location of a point. Each of these numbers indicates the distance between the point and some fixed reference point, called the **origin**. The first number, known as the X value, indicates how far left or right the point is from the origin. The second number, known as the Y value, indicates how far above or below the point is from the origin. The origin has a coordinate of 0, 0.

Longitude and latitude are a special kind of coordinate system, called a **spherical coordinate system**, since they identify points on a sphere or globe. However, there are hundreds of other coordinate systems used in different places around the world to identify locations on the earth. All of these coordinate systems place a grid of vertical and horizontal lines over a flat map of a portion of the earth.

A complete definition of a coordinate system requires the following:

- The projection that is used to draw the earth on a flat map
- The location of the origin
- The units that are used to measure the distance from the origin

GIS Mapping Software

Maptitude Mapping Software gives you all of the tools, maps, and data you need to analyze and understand how geography affects you and your business. Maptitude supports dozens of coordinate systems allowing you to work with data from almost any source.

Types of Coordinate Systems –

Cartesian & Polar Coordinate Systems

Cartesian Coordinate System

As stated above, it uses the concept of mutually perpendicular lines to denote the coordinate of a point. To locate the position of a point in a plane using two perpendicular lines, we use the Cartesian coordinate system. Points are represented in the form of coordinates (x, y) in two-dimension with respect to the x- and y- axes.

The x-coordinate of a point is its perpendicular distance from the y-axis measured along the x-axis, and it is known as abscissa. The y-coordinate of a point is its perpendicular distance from the x-axis measured along the y-axis, and it is known as ordinate. Read More.

Polar Coordinate System

Here, a point is chosen as the pole, and a ray from this point is taken as the polar axis. Basically, we have two parameters, namely angle and radius. The angle ϴ with the polar axis has a single line through the pole measured anti-clockwise from the axis to the line.

The point will have a unique distance from the origin (r). Thus, a point in the polar coordinate system is represented as a pair of coordinates (r, ϴ). The pole is represented by (0, ϴ) for any value of ϴ, where r = 0.

(r, ϴ), (r, ϴ + 2π) and (-r, ϴ + π) are all polar coordinates for the same point.

The distance from the pole is called the radial coordinate, radial distance or simply radius and the angular coordinate, polar angle or azimuth.

Consider the figure below that depicts the relationship between polar and cartesian coordinates.

$X = r \cos \Theta$ and $y = r \sin \Theta$

$r = (x^2 + y^2)^{½}$ and $\tan \Theta = (y/x)$

ypes of Coordinate Systems

A **coordinate system** is used to determine each point uniquely in a plane. The study of the coordinate system led to innovations and research in the field of coordinate geometry. This study was initially developed by the French Philosopher and Mathematician *Rene Descartes*.

We know that a number line is a line that is used to represent the integers in both positive and negative directions from the origin. Rene Descartes invented the idea of placing two such lines perpendicular to each other on a plane and locating points on the plane by referring them to these lines. Though the two perpendicular lines can be in any direction, we usually choose one line on the vertical axis and the other perpendicular to it on the horizontal axis. In this article, we mainly discuss two types of coordinate systems.

Different Types of Coordinate Systems

There are mainly two types of coordinate systems, as listed below:

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Polar Coordinate System

Here, a point is chosen as the pole, and a ray from this point is taken as the polar axis. Basically, we have two parameters, namely angle and radius. The angle Θ with the polar axis has a single line through the pole measured anti-clockwise from the axis to the line.

The point will have a unique distance from the origin (r). Thus, a point in the polar coordinate system is represented as a pair of coordinates (r, Θ). The pole is represented by $(0, \Theta)$ for any value of Θ, where $r = 0$.

(r, Θ), $(r, \Theta + 2\pi)$ and $(-r, \Theta + \pi)$ are all polar coordinates for the same point.

The distance from the pole is called the radial coordinate, radial distance or simply radius and the angular coordinate, polar angle or azimuth.

Consider the figure below that depicts the relationship between polar and cartesian coordinates.

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$r = (x^2 + y^2)^{½}$ and $\tan \Theta = (y/x)$

The polar equation of a curve consists of points of the form (r, Θ).

In the case of a circle, the general equation for a circle with a centre at **(R, β)** and

radius **a** is $\mathbf{r^2 - 2rR \cos(\Theta - \beta) + R^2 = a^2}$.

Radial lines (those running through the pole) are represented by the equation: $\Theta = \beta$.

Cartesian Formulae for the Plane

Distance between two points

The distance between two points of the plane (x_1, y_1) and (x_2, y_2) is given by

$$d = [(x_2 – x_1)^2 + (y_2 – y_1)^2]^{1/2}$$

In the case of a three-dimensional system, the distance formula between the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$d = [(x_2 – x_1)^2 + (y_2 – y_1)^2 + (z_2 – z_1)^2]^{1/2}$$

Representation of a vector

In two dimensions, the vector from the origin to the point with the Cartesian coordinates (x, y) can be written as r = x**i** + y**j,** where **i** = (1,0) and **j** = (0,1) are unit vectors in the direction of the x-axis and y-axis, respectively.

In the case of three dimensions, we will have r = x**i** + y**j** + z**k**, where k = (0,0,1) is the unit vector in the direction of the z-axis.

Also, Read:

Three-dimensional Geometry

Distance formula

Example Problems

Example 1: On graph paper, plot the point where Ajay has kept the lamp. He describes the situation as: the lamp is kept on a table which is taken as the reference. The lamp has a height of 2 units and is kept at the edge of the beginning end of the table.

Solution: As per the statement given in the question, the reference table is to be considered as the origin (0, 0). The position of the lamp, as stated above, clearly points out to be on the coordinate (0, 2). Thus, the point marked A is the position of the lamp with respect to the table reference marked as B. The following graph illustrates this.**Example 2:** Find the distance between the two points (4,5,6) and (9,8,2) in the x-y plane.

Solution: $d = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$

Given: $(x_1, y_1, z_1) = (4,5,6)$ and $(x_2, y_2, z_2) = (9, 8, 2)$

Therefore, $x_2 - x_1 = 9 - 4 = 5$.

Similarly, $y_2 - y_1 = 8 - 5 = 3$ and $z_2 - z_1 = 2 - 6 = -4$

Hence, $d^2 = 5^2 + 3^2 + (-4)^2$

or, $d = \sqrt{50}$ units.